

24.4 A 512ch 10b Source-Driver IC with Current-Mode Auto-Zeroing Scheme Achieving 1.4mV DVO and 600ns 1-Horizontal Time for Mobile OLED Displays

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Abstract

This paper presents a 10b source-driver IC (SD-IC) for ultra-fast, uniform-luminance mobile OLED displays. A current-mode auto-zeroing (CM-AZ) scheme transparently cancels output-buffer offset—lowering DVO with zero timing penalty—while simultaneously suppressing

AC-coupling errors in the LSB-stacked DAC. Fabricated in 180nm CMOS with an 11.37μm channel pitch, the 512ch SD-IC achieves a 600ns 1-H time and a 1.4mV DVO, successfully demonstrated on an LED display panel.

The growing demand for mobile OLED displays with ultra-high frame rates and superior luminance uniformity drives advances in source-driver ICs (SD-ICs) [1,2]. Pushing the frame rate beyond 240Hz—well above the legacy 60Hz—in high-resolution panels shortens the one-horizontal (1-H) interval, defined as 1-H time = 1 / (frame rate × number of scan lines). Within this 1-H period, the SD-IC must complete digital-to-analog conversion (DAC) and program each OLED pixel through its buffer amplifier. Display uniformity, meanwhile, depends on minimizing the deviation of voltage outputs (DVO) across hundreds of source channels—a metric dominated by the offset-voltage (V_{OS}) dispersion of the output buffers.

Frame-by-frame chopping [3–6] has been explored to lower DVO by alternately inverting the V_{OS} polarity every frame; however, modern variable-refresh-rate (VRR) displays can throttle the frame rate down to 1Hz to save power [7–9], and at such ultra-low rates, the luminance ripple become visible flicker. Conventional voltage-mode auto-zeroing (AZ) [10–13] cancels the buffer's V_{OS} by sampling it onto a capacitor, but it suffers two major drawbacks: (1) the dedicated V_{OS} -sampling phase consumes precious 1-H timing margin, and (2) the AZ capacitor is directly coupled to the output. This coupling allows sample-and-hold (S/H) errors to appear at the output, ultimately limiting the attainable DVO. Foreground calibration [14,15] is an option, but requires a lengthy start-up, is constrained by the calibration resolution, and cannot track temporal V_{OS} drift during normal operation.

This paper proposes a current-mode auto-zeroing (CM-AZ) technique. Unlike classical AZ, the CM-AZ removes buffer offsets transparently during normal data driving with zero timing overhead. Leveraging this scheme, we implemented a 512-channel, 10b SD-IC that achieves both a low DVO and a fast 1-H time.

The top-left of Fig. 24.4.1 shows a simplified model of a typical unity-gain buffer. The 1st stage is a folded-cascode amplifier that sets the transconductance (G_m) and boosted output resistance (r_{o1}). It comprises a differential input-pair (DIP) biased by a tail current of $2 \cdot I_B$, bias current sources of ($I_B + I_S$), and a cascode stage biased by $2 \cdot I_S$ that acts as a current conveyor. The 2nd-stage amplifier (-A) provides additional gain. An input-referred offset voltage (V_{OS}), caused primarily by transistor mismatch in the DIP, results in a final output of $V_O = V_{IN} + V_{OS}$, rather than the ideal $V_O = V_{IN}$. Conceptually partitioning the combined bias ($I_B + I_S$) into individual I_B and I_S sources reveals a zero-current bridge (ZCB) between them (top-right of Fig. 24.4.1). This structure presents a key opportunity to correct the DIP's V_{OS} . If an offset current (I_{OS}) that matches V_{OS} (as $I_{OS} = G_m \cdot V_{OS}$) could be intentionally injected into the left side of the ZCB, the offset would be cancelled, yielding the ideal $V_O = V_{IN}$. In a steady state, the ZCB carries no DC current and thus behaves like a virtual open circuit. These observations—that a purposefully introduced I_{OS} can compensate V_{OS} , and that the ZCB virtually isolates the V_{OS} -corrected DIP from the cascode stage—led to the conception of our proposed current-mode auto-zeroing (CM-AZ) technique.

The bottom of Fig. 24.4.1 presents the CM-AZ architecture, which transparently cancels V_{OS} in the background while the buffer drives V_O . The original I_B sources are replaced by current-mode S/Hs positioned to the left side of the ZCB, and an auxiliary DIP is added to its right side. Operation proceeds in two overlapping phases. During the initial drive—or sampling—interval (T_{SAMP}), as new data arrives at V_{IN} , the auxiliary DIP steers V_O to slew toward V_{IN} . Concurrently, the main DIP is isolated from the bridge with both its inputs shorted to V_{IN} , allowing it to produce the V_{OS} -matched offset current $I_{OS} (= G_m \cdot V_{OS})$. The S/H then samples a current of $I_B \pm I_{OS}$. After T_{SAMP} , the S/H switches to hold mode, and control of the buffer is returned to the main DIP to complete the settling of V_O . Because the sampled $\pm I_{OS}$ actively cancels V_{OS} from the main DIP, any offset component is prevented from propagating into the cascode stage even at $V_O = V_{IN}$. Consequently, as V_O settles to V_{IN} , it remains free of offset, and the bridge current becomes zero ($I_{ZCB} = 0$). The DC-floating nature of the ZCB enables a seamless hand-off between the (temporary-drive) auxiliary DIP and the (now offset-compensated) main DIP. This achieves in-operation (“zero-wait”) offset cancellation without a dedicated sampling slot, meaning the CM-AZ technique incurs no speed penalty.

Figure 24.4.2 depicts the schematic of the CM-AZ buffer amplifier. During T_{SAMP} ($t_0 < t < t_1$), when $\langle \Phi_S, \Phi_H \rangle = \langle high, low \rangle$, the negative input node of the main DIP is connected to V_{IN} via MUX₁. Concurrently, the current-mode S/H (M_{SH} and C_{SH}) samples the main DIP's currents, $I_{IN1} (= I_B + I_{OS})$ and $I_{IN2} (= I_B - I_{OS})$. For a rising step at V_{IN} , the auxiliary DIP temporarily supplies slewing currents, $I_{SUM+} (= 2 \cdot I_B - I_B)$ and $I_{SUM-} (= 0 - I_B)$, to the folded-

cascode stage, driving the initial positive transition of V_O . After T_{SAMP} (during $t_1 < t < t_2$), with $\langle \Phi_S, \Phi_H \rangle = \langle low, high \rangle$, MUX_{2,3} hand control of the buffer amplifier back from the auxiliary DIP to the main DIP while the S/H pair holds the sampled currents. Because V_O is still slewing, the main DIP now delivers $I_{SUM+} = I_B - I_{OS}$ and $I_{SUM-} = -(I_B - I_{OS})$ to the following stage. The nearly unchanged magnitude of I_{SUM+} before and after the DIP hand-over ensures a smooth transition. As the buffer amplifier enters its fine-settling phase, the I_{OS} term stored in the S/H cancels (absorbs) the main DIP's offset V_{OS} , driving the bridge current I_{ZCB+} toward true-zero and forcing V_O to accurately track V_{IN} , free of V_{OS} .

In each 1-H cycle, the S/H circuits (M_{SH} and C_{SH}) capture a nearly constant $I_B \pm I_{OS}$, as V_{OS} changes negligibly from one cycle to the next. Thus, a long sampling time (T_{SAMP}) is not required to achieve high accuracy, making the CM-AZ offset cancellation insensitive to the exact T_{SAMP} . Because a high sampling rate—set by $g_{m,SH}/C_{SH}$ —is not needed, M_{SH} can be designed with a low $g_{m,SH}$. A key benefit of this low $g_{m,SH}$ is its ability to effectively attenuate error voltages on C_{SH} , which in turn allows C_{SH} to be kept small ($\leq 40fF$). Moreover, since the V_{OS} is cancelled from the instantaneous V_{IN} level, the CM-AZ continuously tracks V_{OS} variations caused by DC shifts in V_{IN} —an advantage over foreground-calibrations. The auxiliary DIP adds only negligible overhead; with the offset effectively suppressed, both the main and auxiliary DIPs can be designed with minimum area. Although CM-AZ cannot correct the mismatch current between M_{P1} (M_{N1}) and M_{P2} (M_{N2}) of the folded-cascode stage, that mismatch is attenuated by the main DIP's G_m . It appears only as a negligible input-referred offset—less than 2% of the total V_{OS} in our design.

Figure 24.4.3 describes the in-operation LSB-stacking DAC scheme. In our SD-IC design, we adopt the LSB-stacking DAC of [12]: a 6b resistor-string DAC (R-DAC) first performs a coarse conversion, producing V_H and $V_L (= V_{MSB})$. Subsequently, a 4b switched-capacitor (SC) DAC interpolates V_{LSB} between V_H and V_L and then “stacks” this fine voltage onto V_{MSB} , resulting in the 10b $V_O = V_{MSB}$ (6b) + V_{LSB} (4b). However, [12] suffers from an AC-coupling error (top-left of Fig. 24.4.3). If V_{MSB} is driven to the output V_O while stacking V_{LSB} via C_{LSB} , the parasitic C_{IN-} of the buffer's negative input (V_{IN-}) introduces an error voltage, $V_E [= C_{IN-}/C_{LSB} \cdot (V_{MSB} - V_X)]$, where V_X is the buffer's V_{IN} voltage at the V_{LSB} -stacking instant ($t = t_1$). To avoid V_E , driving V_{MSB} to V_O and stacking V_{LSB} must be time-separated, lengthening the total DAC interval. The CM-AZ buffer can remove this constraint (right of Fig. 24.4.3). During the I_{OS} -sampling phase ($t_0 < t < t_1$), MUX₁ (with $\Phi_H = low$) ties the main DIP's V_{IN-} directly to $V_{IN+} (= V_{MSB})$; this pre-charges C_{IN-} to V_{MSB} (i.e., $V_X = V_{MSB}$) prior to the V_{LSB} -stacking event, thereby suppressing V_E and obviating time-separation of the two operational phases. Thus, the CM-AZ buffer inherently enables faster operation of the LSB-stacking DAC.

The top of Fig. 24.4.4 shows the architecture of the 512-channel SD-IC. Each channel processes serial data through latches and level shifters, generating a 10b voltage (V_O) using a combined 6b R-DAC and 4b LSB-stacking SC-DAC. To drive the highly capacitive data lines of a high-resolution OLED panel, the CM-AZ buffer amplifier is augmented with a high-slew-rate (HSR) assist stage (top-right of Fig. 24.4.4). With V_{IN} and V_O as inputs, the HSR circuit detects rising or falling transitions by comparing the transient current via $M_{P(N)<8>6}$, with a reference I_{CMP} . Once the detection threshold is exceeded, switch $M_{N(P)9}$ turns on, allowing an assist current I_{SR} —much larger than the nominal bias I_B —to be pulled from or pushed to the buffer's intermediate node $V_{X(Y)}$, thereby accelerating the output slew rate.

The 512-channel 10b SD-IC was fabricated in a 180nm CMOS (Fig. 24.4.7). The bottom-left of Fig. 24.4.4 presents the output (V_O) waveforms measured with various 10b input codes ($D<9:0>$) under a display panel load of 30kΩ and 30pF. The HSR buffer achieves a slew rate of 31.6V/μs—a 4.78× improvement over the non-HSR setup—without increasing the quiescent current ($I_Q = 1.7μA$). The V_O settled with the LSB-stacking DAC exhibits no AC-coupling error ($V_E = 0$). By contrast, if AC-coupling issue were not addressed, V_O would be severely distorted. This work achieves the minimum 1-H time of 600ns while guaranteeing 10b accuracy. The bottom-center of Fig. 24.4.4 confirms that the CM-AZ incurs no speed penalty; the V_O -settling curves with and without CM-AZ are essentially identical, demonstrating transparent offset cancellation during the buffer's normal operation. The bottom-right of Fig. 24.4.4 shows that the CM-AZ performance remains stable for different offset-sampling windows ($T_{SAMP} = 200ns$ and 250ns).

The top of Fig. 24.4.5 presents demonstration photos from a 128×128 red/green LED display panel driven by the SD-IC. As confirmed by the middle of Fig. 24.4.5, the CM-AZ effectively suppresses offsets. Among 128 channels, the maximum DVO of 1.4mV was measured, which is a 7.7× reduction over the non-CM-AZ version, achieved under identical silicon area, 1-H time, and power budget constraints. The bottom of Fig. 24.4.5 shows the measured linearity of the 10b DAC. The maximum DNL was 0.27LSB. Applying 7 voltage-taps (used for gamma-correction) to the global 6b R-string improved the maximum INL to 0.37LSB. Figure 24.4.6 presents a performance comparison to prior art and the 1-channel layout. Compared to prior art, the proposed chip achieves the smallest DVO (1.4mV) and the fastest 1-H time (600ns), all while maintaining a competitive size and a wide rail-to-rail output range.

Acknowledgement:

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References:

- [1] Y.-R. Jo *et al.*, "An OLED Display Driver IC with High-Gain Fast-Slew Circuit and On-the-Fly Self-Repair Technique for High-Resolution Display," *J. SID*, vol. 32, no. 5, pp. 415–425, May 2024. <http://doi.org/10.1002/jsid.1307>
- [2] Y. Park *et al.*, "A 10-bit Source-Driver IC with Charge-Modulation DAC for Enhanced Frame-Rate Mobile OLED Displays," *IEEE JSSC*, vol. 59, no. 11, pp. 3511–3524, Nov. 2024. <http://doi.org/10.1109/JSSC.2024.3442248>
- [3] J.-S. Kang *et al.*, "10-bit Driver IC Using 3-bit DAC Embedded Operational Amplifier for Spatial Optical Modulators (SOMs)," *IEEE JSSC*, vol. 42, no. 12, pp. 2913–2922, Dec. 2007. <http://doi.org/10.1109/JSSC.2007.908690>
- [4] H.-M. Lee *et al.*, "An Area and Power Efficient Interpolation Scheme Using Variable Current Control for 10-Bit Data Drivers in Mobile Active-Matrix LCDs," *IEEE Trans. Consum. Electron.*, vol. 65, no. 2, pp. 253–262, May 2019. <http://doi.org/10.1109/TCE.2019.2900512>
- [5] J.-S. Na *et al.*, "A Highly Linear 10-Bit DAC of Data Driver IC Using Source Degeneration Load for Active Matrix Flat-Panel Displays," *IEEE TCAS-II*, vol. 67, no. 11, pp. 2312–2316, Nov. 2020. <http://doi.org/10.1109/TCSII.2020.2972370>
- [6] K. Ryu *et al.*, "A Source-Driver IC Including Power-Switching Fast-Slew-Rate Buffer and 8Gb/s Effective 3-Tap DFE Receiver Achieving 4.9mV DVRMS and 17V/μs Slew Rate for 8K Displays and Beyond," *ISSCC*, Feb. 2023, pp. 382–383. <http://doi.org/10.1109/ISSCC42615.2023.10067592>
- [7] Y. Kim *et al.*, "A Highly Uniform Luminance and Low-Flicker Pixel Circuit and Its Driving Methods for Variable Frame Rate AMOLED Displays," *IEEE Access*, vol. 11, pp. 74301–74311, July 2023. <http://doi.org/10.1109/ACCESS.2023.3296787>
- [8] Y. Kim *et al.*, "A Pixel Circuit with Improved Luminance Uniformity and Flicker for AMOLED Displays With a Wide VRR Range of 15 Hz to 360 Hz," *IEEE Electron Device Lett.*, vol. 46, no. 4, pp. 600–603, April 2025. <http://doi.org/10.1109/LED.2025.3535717>
- [9] D. Sim *et al.*, "Improvement of Flicker Phenomenon at Low Frequencies in AMOLED Displays by Applying Compensation Scheme of Variable Reset Voltage," *Sci. Rep.*, vol. 15, no. 1, April 2025. <http://doi.org/10.1038/s41598-025-96847-1>
- [10] C.-W. Lu, "A Rail-To-Rail Class-AB Amplifier with an Offset Cancellation for LCD Drivers," *IEEE JSSC*, vol. 44, no. 2, pp. 525–537, Feb. 2009. <http://doi.org/10.1109/JSSC.2008.2010995>

- [11] M.-W. Hsu and C.-L. Chen, "A Cost-Effective Offset Cancellation Structure for LCD Source Driver," *ISCAS*, June 2014, pp. 2317–2320. <http://doi.org/10.1109/ISCAS.2014.6865635>
- [12] G.-W. Lim *et al.*, "An Area-Efficient 10-Bit Source-Driver IC With LSB-Stacked LV-to-HV-Amplify DAC for Mobile OLED Displays," *IEEE JSSC*, vol. 58, no. 11, pp. 3164–3178, Nov. 2023. <http://doi.org/10.1109/JSSC.2023.3289503>
- [13] J. Ahn *et al.*, "A Fully Nonlinear Compact 10b Source Driver with Low-Voltage Gamma Slope DAC and Data/Phase Dependent Current Modulation Achieving 2411μm²/Channel for Mobile OLED Displays," *ISSCC*, Feb. 2024, pp. 434–435. <http://doi.org/10.1109/ISSCC49657.2024.10454504>
- [14] C. Kim *et al.*, "Effective 10-Bit OLED Driver IC with 11-Bit DAC, Double Capacitor-Coupled Adder, and Offset Calibration for Enhanced Panel Driving," *J. SID*, vol. 33, no. 5, pp. 644–652, April 2025. <http://doi.org/10.1002/jsid.2068>
- [15] J. Oh *et al.*, "A 10b Source-Driver IC with All-Channel Automatic Offset Calibration and Slew-Rate-Enhanced Amplifier Achieving 2273pm²/Channel and 1.9mV DVO for 6285-PPI OLED-on-Silicon Displays," *ISSCC*, Feb. 2025, pp. 126–127. <http://doi.org/10.1109/ISSCC49661.2025.10904712>

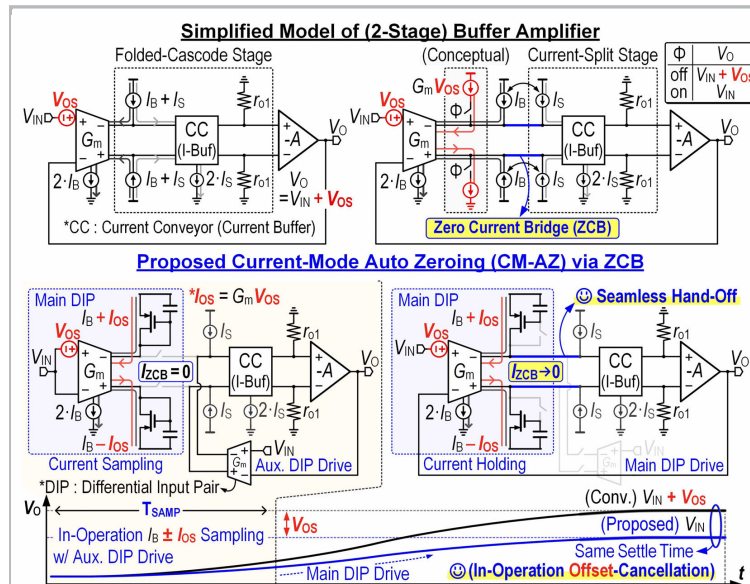


Figure 24.4.1: Simplified model of a conventional buffer using a two-stage amplifier (top) and the proposed current-mode auto-zeroing (CM-AZ) technique (bottom).

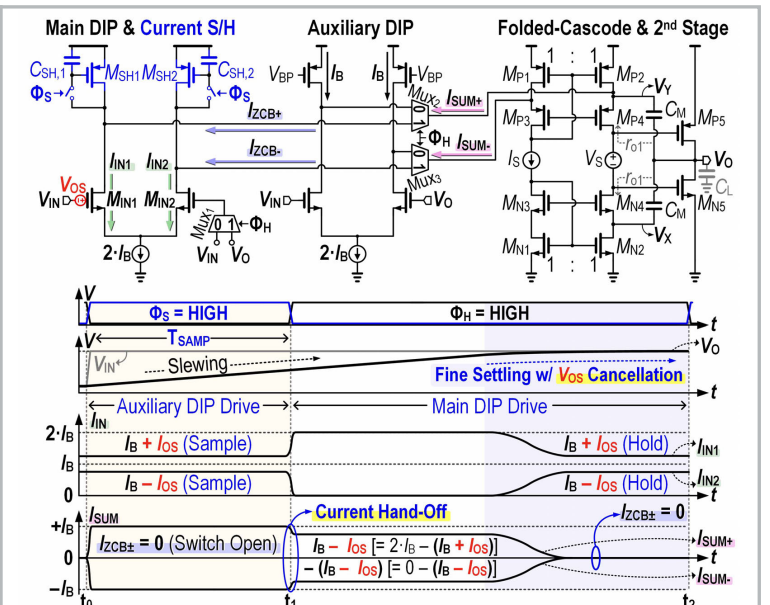
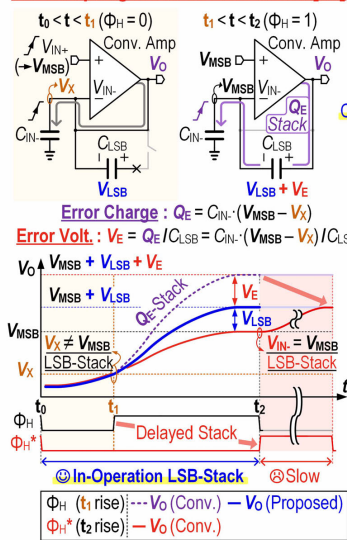
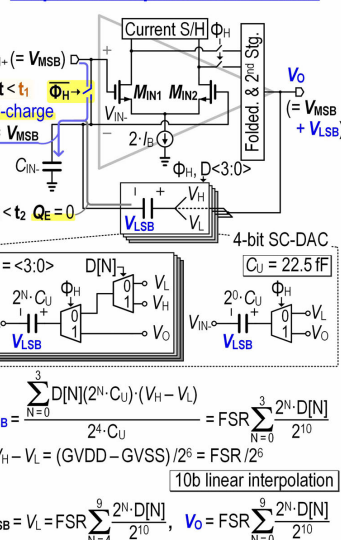


Figure 24.4.2: Schematic of the CM-AZ buffer amplifier and its key timing diagram.

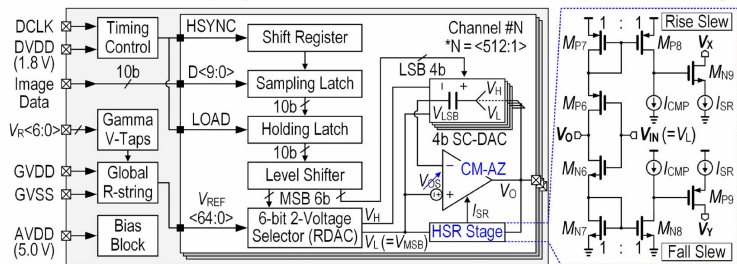
AC-Coupling Issue in LSB-Stack [12]



Proposed In-Operation LSB-Stack



Architecture of the proposed 512-channel 10-bit SD-IC



Measured V_{OUT} Waveforms

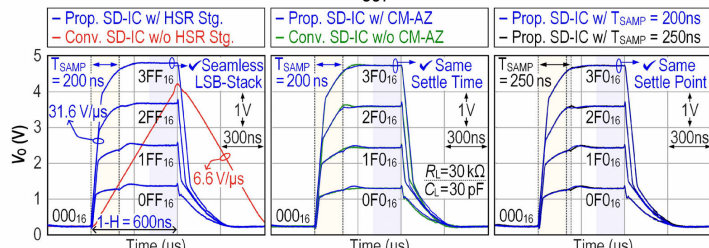


Figure 24.4.3: Mitigation of the AC-coupling error in LSB-stacking SC-DAC using the inherent C_{IN} -pre-charge of the CM-AZ technique.

Figure 24.4.4: Top architecture of the 512-ch 10b SD-IC (top-left), schematic of the proposed high slew-rate stage (top-right), and measured key waveforms (bottom).

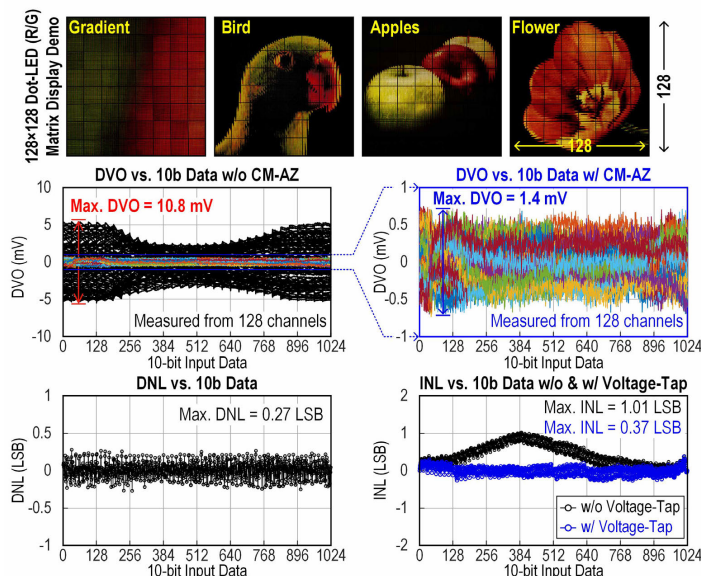
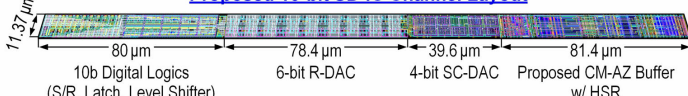


Figure 24.4.5: Photograph of the LED display demonstration driven by the SD-IC, measured deviation of voltage outputs (DVO), and linearity (DNL/INL).

Proposed 10-bit SD-IC Channel Layout



	ISSCC'23 [6]	JSSC'23 [12]	JSSC'24 [2]	ISSCC'25 [15]	This Work
CMOS Process	180-nm	130-nm	180-nm	65-nm	180-nm
Gray Scale (DAC Architecture)	— (R-DAC)	10-bit (SC-multiplier + LSB stack-up)	10-bit (QM-DAC)	10-bit (DAC-embedded Amplifier)	10-bit (SC-DAC)
Output Range	0.2 to 17.8 V	0.3 to 4.5 V	0.25 to 4.85 V	2.1 to 3.1 V*	0.2 to 4.8 V
Static Current (I_{O}/ch)	19 μA	1.8 μA	2.4 μA	5.4 μA	1.7 μA
DNL/INL	—	0.39/0.9 LSB	0.21/0.41 LSB	0.27/0.59 LSB	0.27/0.37 LSB
Offset Cancellation Method	Per-Frame Chopping	Auto-Zeroing	None	Foreground Calibration	CM-AZ
Overhead	Visible Flicker	Long 1-H	High DVO	In-situ V_{OS} Drift	None
Max. DVO	4.9 mV	4.82 mV	11.5 mV	1.9 mV	1.4 mV
1-Horizontal Time***	1** μs (1.67 x)	8.2 μs (13.67 x)	1.5 μs (2.5 x)	0.69 μs (1.15 x)	0.6 μs (1 x)
Slew Rate @ R_L & C_L	17 V/ μs @ 5 k Ω & 350 pF	8.1** V/ μs @ 30 k Ω & 30 pF	22 V/ μs @ 3 k Ω & 100 pF	6.8 V/ μs @ 10 k Ω & 10 pF	31.6 V/ μs @ 30 k Ω & 30 pF
DAC + Buffer Area per Channel	10 x 560** μm^2 (2.47 x)	16 x 144 μm^2 (1.02 x)	15 x 221 μm^2 (1.46 x)	16 x 123.8 μm^2 (0.87 x)	11.37 x 199 μm^2 (1 x)

* Output range for OLEDs application ** Estimation from die micrograph or measured waveforms

*** 1-H time = The time required for the analog output to settle within 10b accuracy after a digital input

Figure 24.4.6: Performance summary with 1-channel layout of the SD-IC.

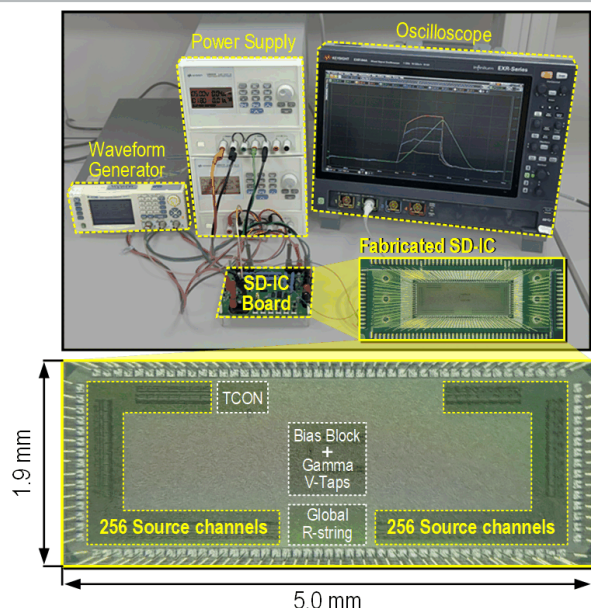


Figure 24.4.7: Die micrograph of the fabricated 512-ch source-driver IC (SD-IC).